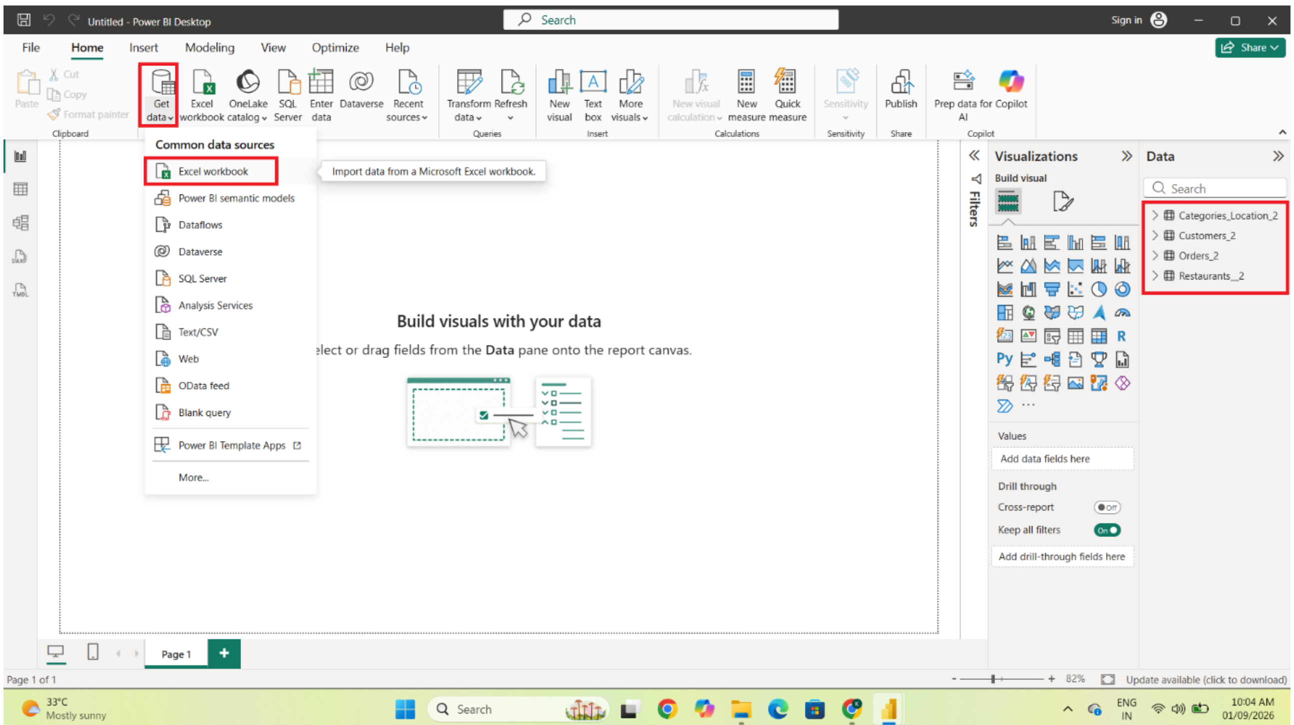
Data Visualization Using Power BI

DAX Measures & Calculated Column

Import the source tables

Steps: Open Power BI Desktop and select Home → Get Data → Excel. Choose the project workbook, select all four tables (Restaurants, Orders, Customers, and Categories_Locations), and load them into Power BI. Verify that all four tables appear in the Data pane.

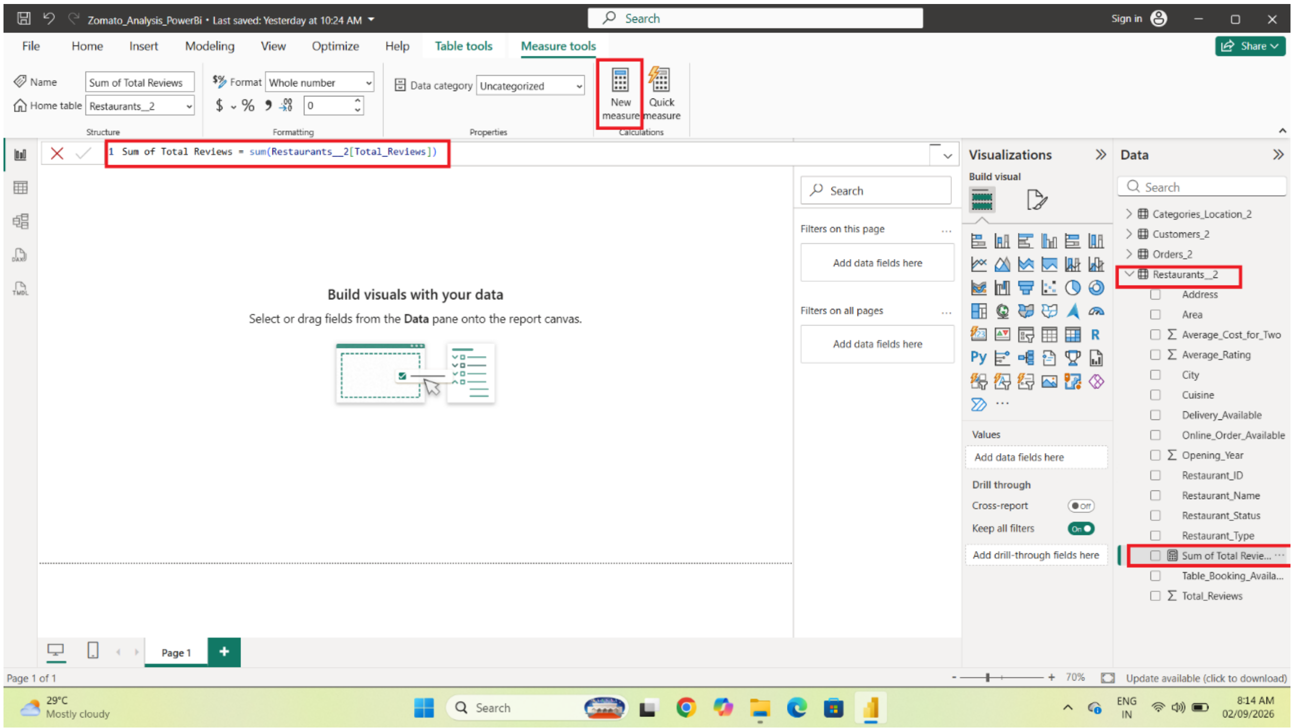
All four tables are imported into Power BI.



Create the Total Reviews measure

Steps: Select the Restaurants table in the Data pane. Choose Modeling → New measure. Enter the Total Reviews measure and press Enter. Check that the measure is created under the Restaurants table.

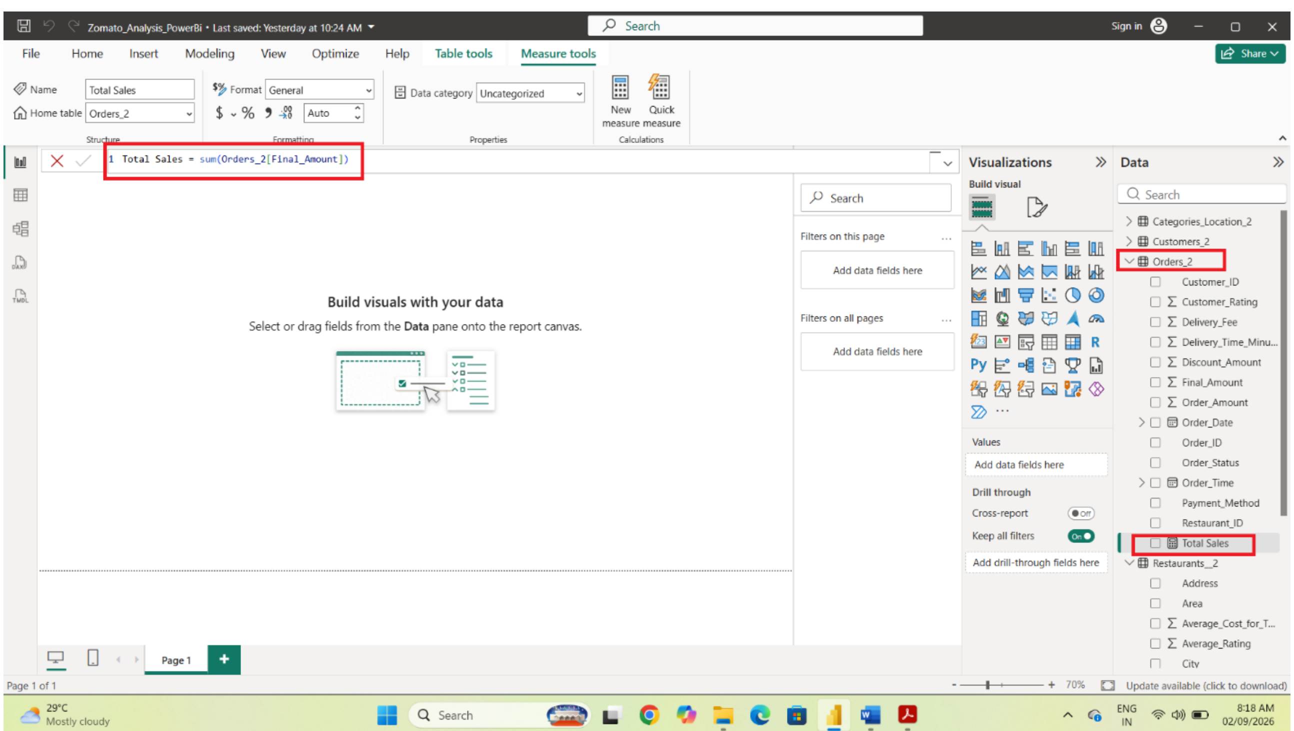
Create the Total Reviews measure from the Restaurants table.



Create the Total Sales measure

Steps: Select the Orders table, choose Modeling → New measure, and create Total Sales by summing the Final Amount/Final_Order_Amount field. Press Enter and verify the measure in the Data pane.

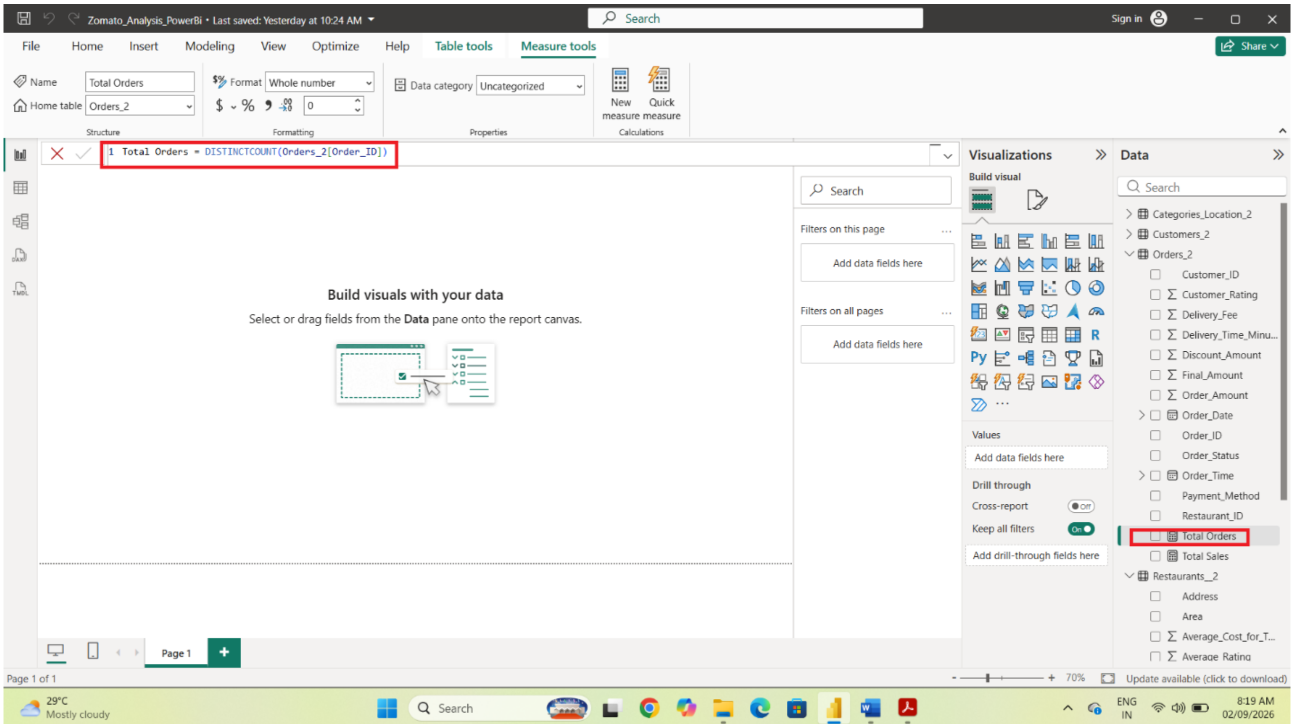
Create a measure to calculate total sales from the order final amount.



Create the Total Orders measure

Steps: Select the Orders table and choose Modeling → New measure. Create Total Orders using a distinct count of Order_ID so that duplicate order records do not inflate the order count. Press Enter.

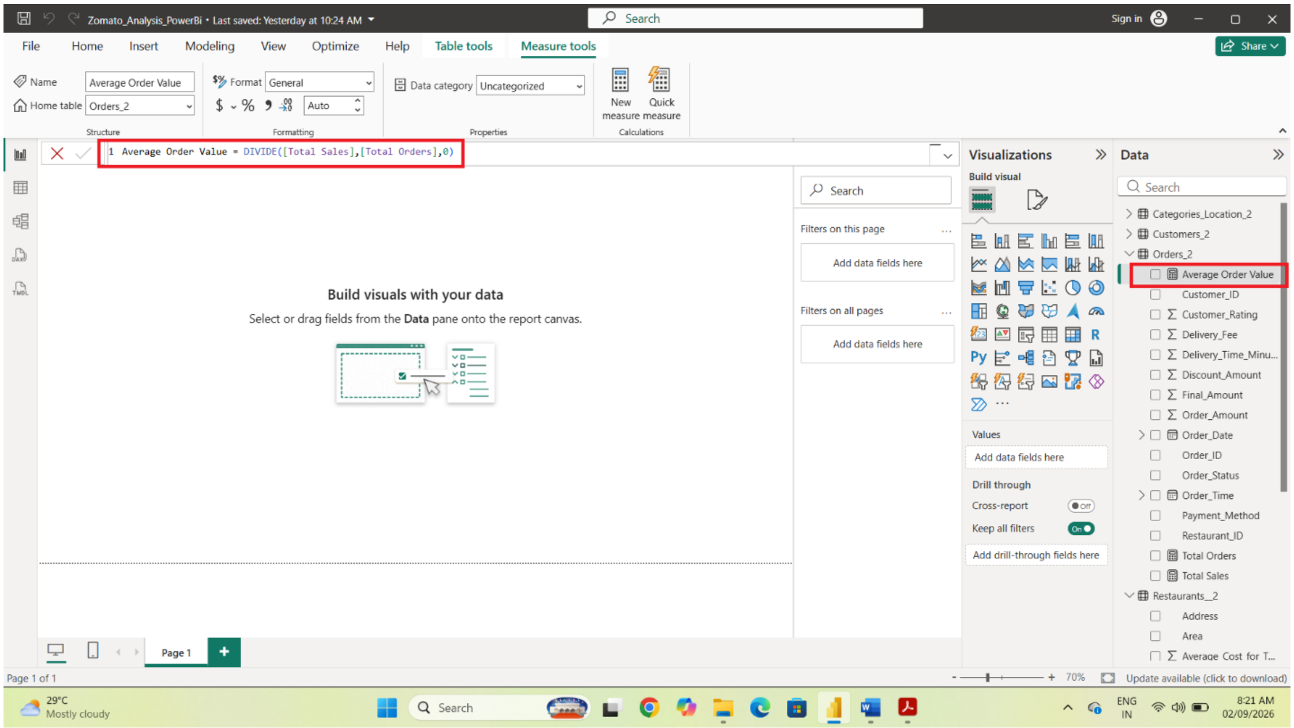
Create a distinct-count measure for Order_ID.



Create the Average Order Value measure

Steps: Select Modeling → New measure and create Average Order Value by dividing Total Sales by Total Orders. Use DIVIDE so the calculation is protected against a zero denominator.

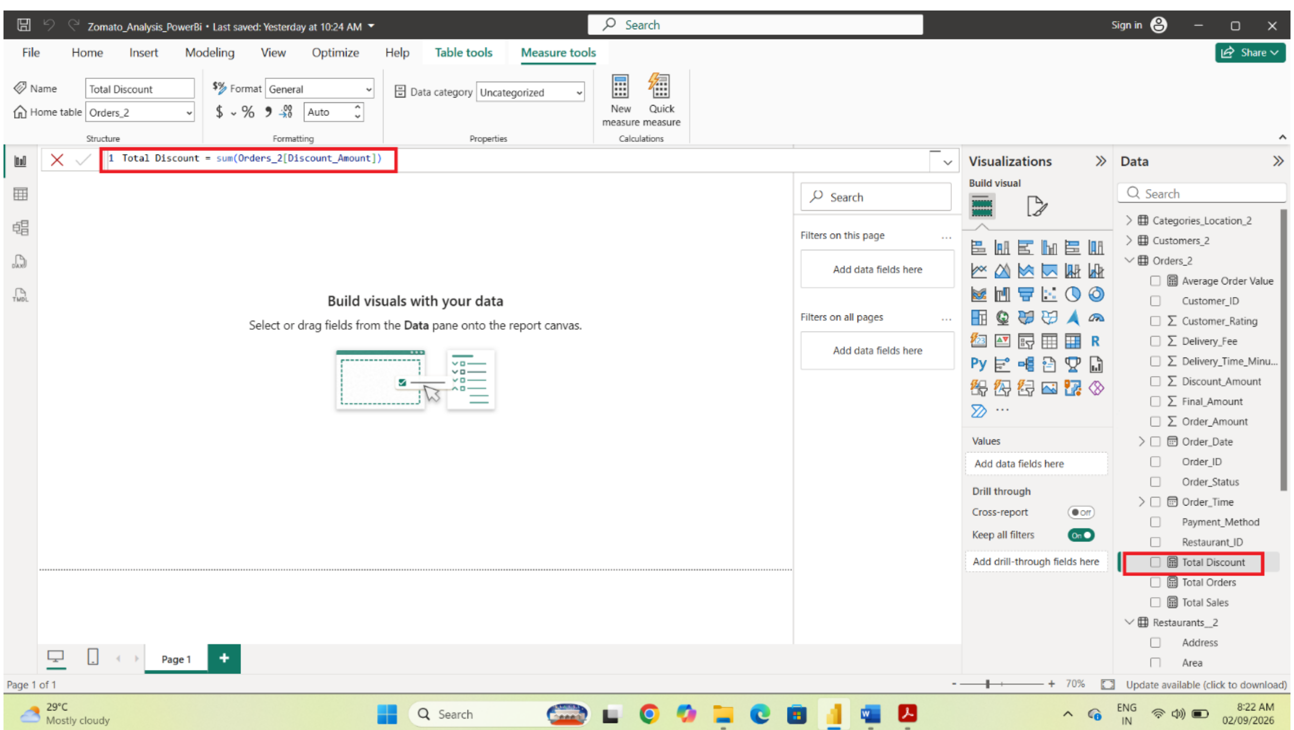
Create Average Order Value from Total Sales and Total Orders.



Create the Total Discount measure

Steps: Select the Orders table → Modeling → New measure. Create Total Discount by summing the Discount Amount/Discount_Amount field. Press Enter and verify the result.

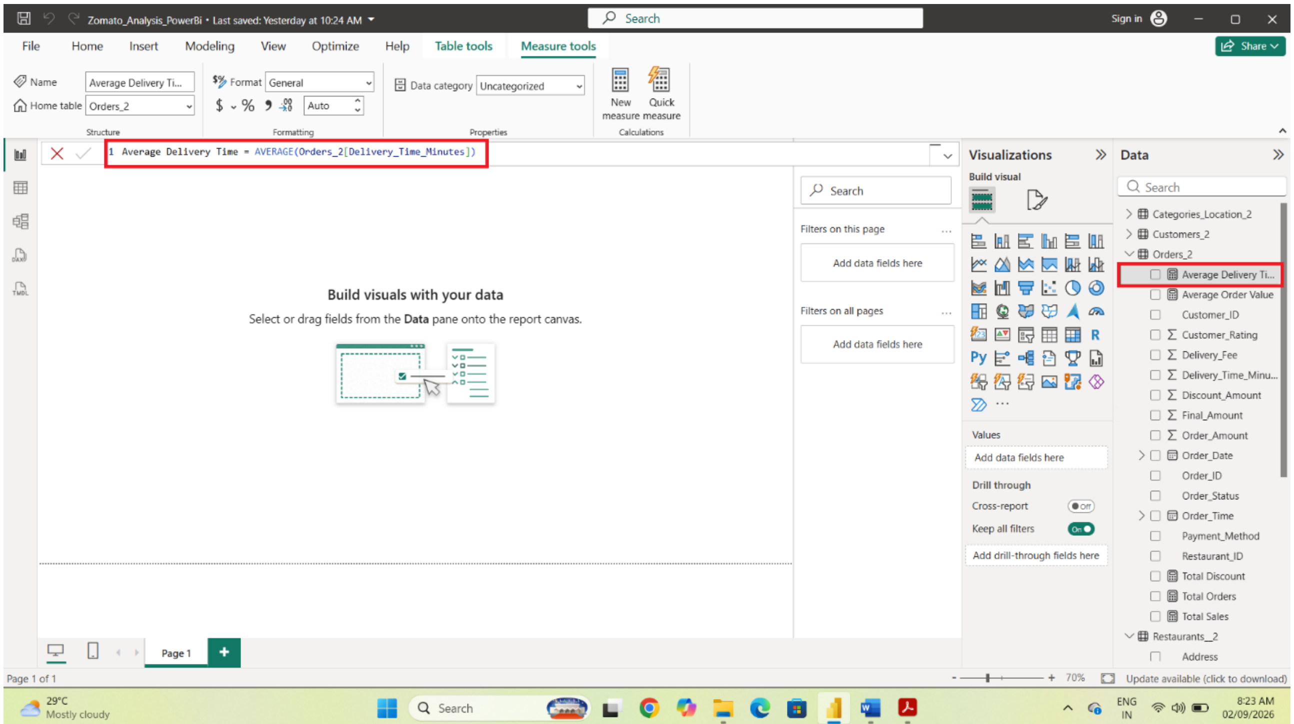
Create a measure to calculate the total discount provided.



Create the Average Delivery Time measure

Steps: Select the Orders table → Modeling → New measure. Create Average Delivery Time using the average of Delivery Time Minutes. Format the result appropriately for reporting.

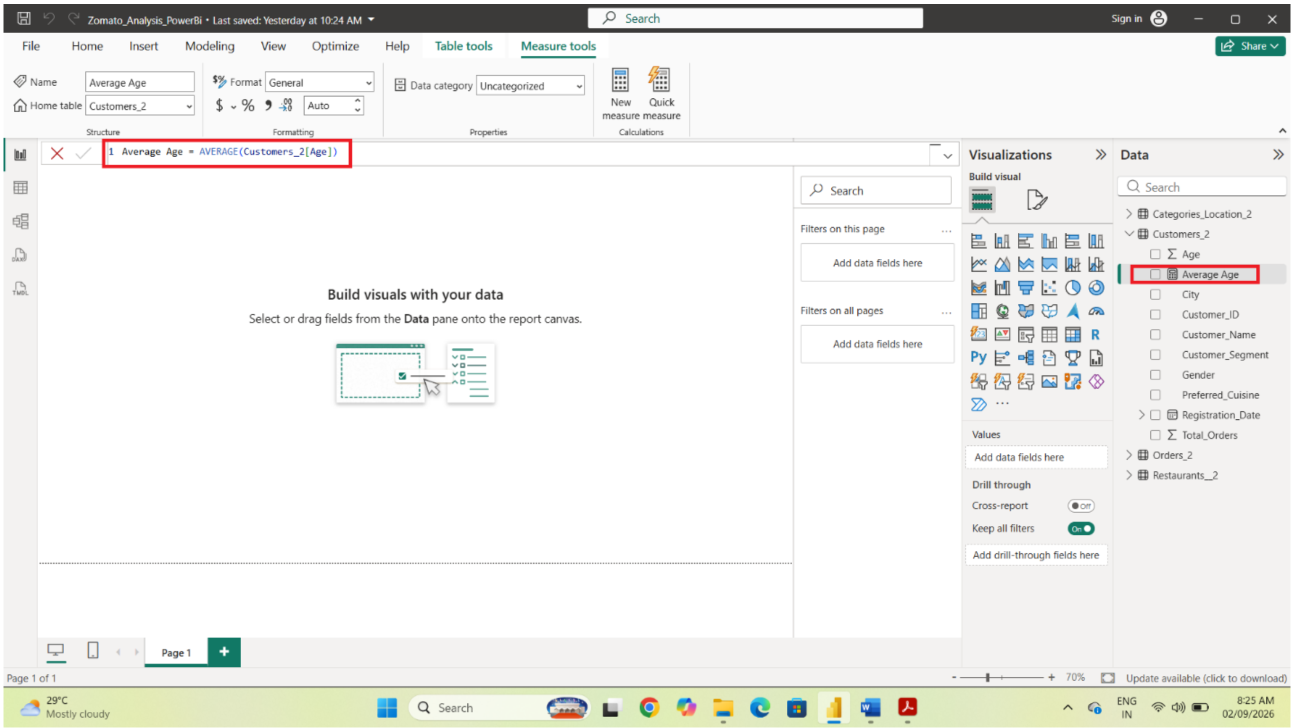
Create the average delivery-time measure.



Create the Average Age measure

Steps: Select the Customers table → Modeling → New measure. Create Average Age using the average of the Age field. Press Enter and verify the measure.

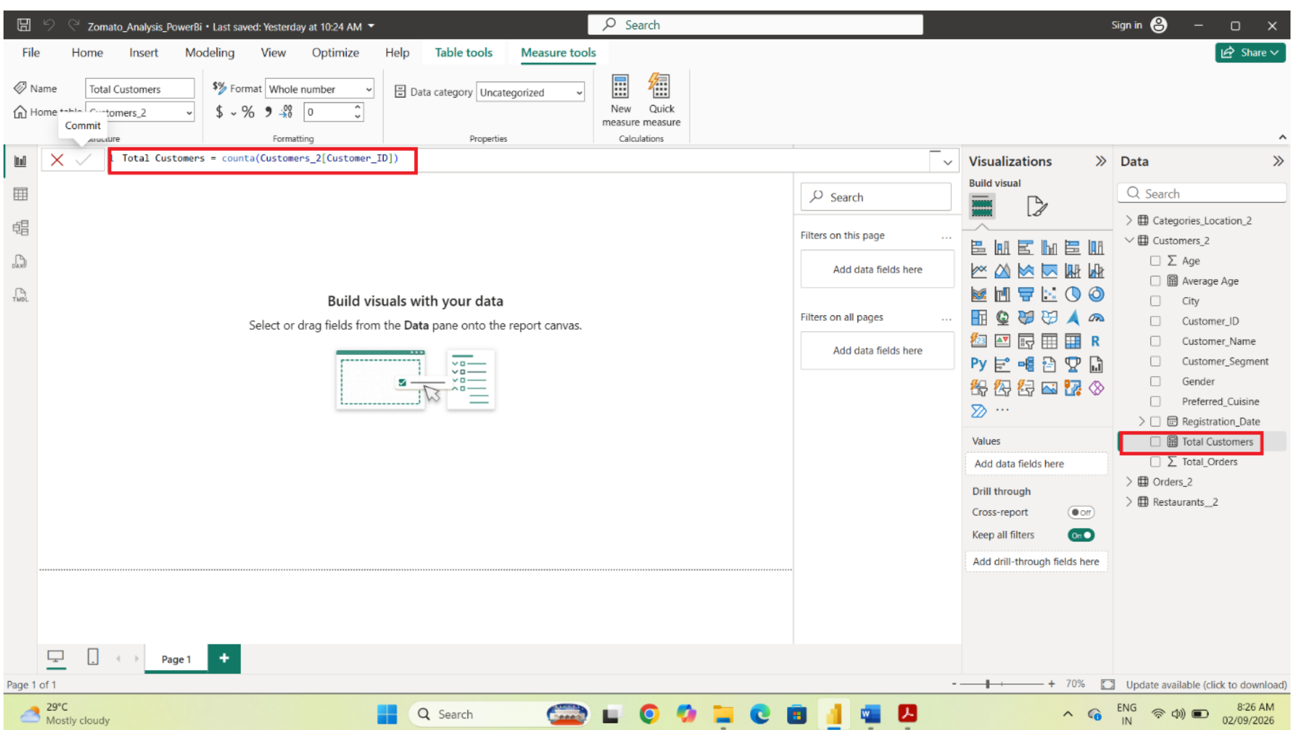
Create a measure to calculate the average customer age.



Create the Total Customers measure

Steps: Select the Customers table → Modeling → New measure. Create Total Customers using a distinct count of Customer_ID. Press Enter and verify the measure.

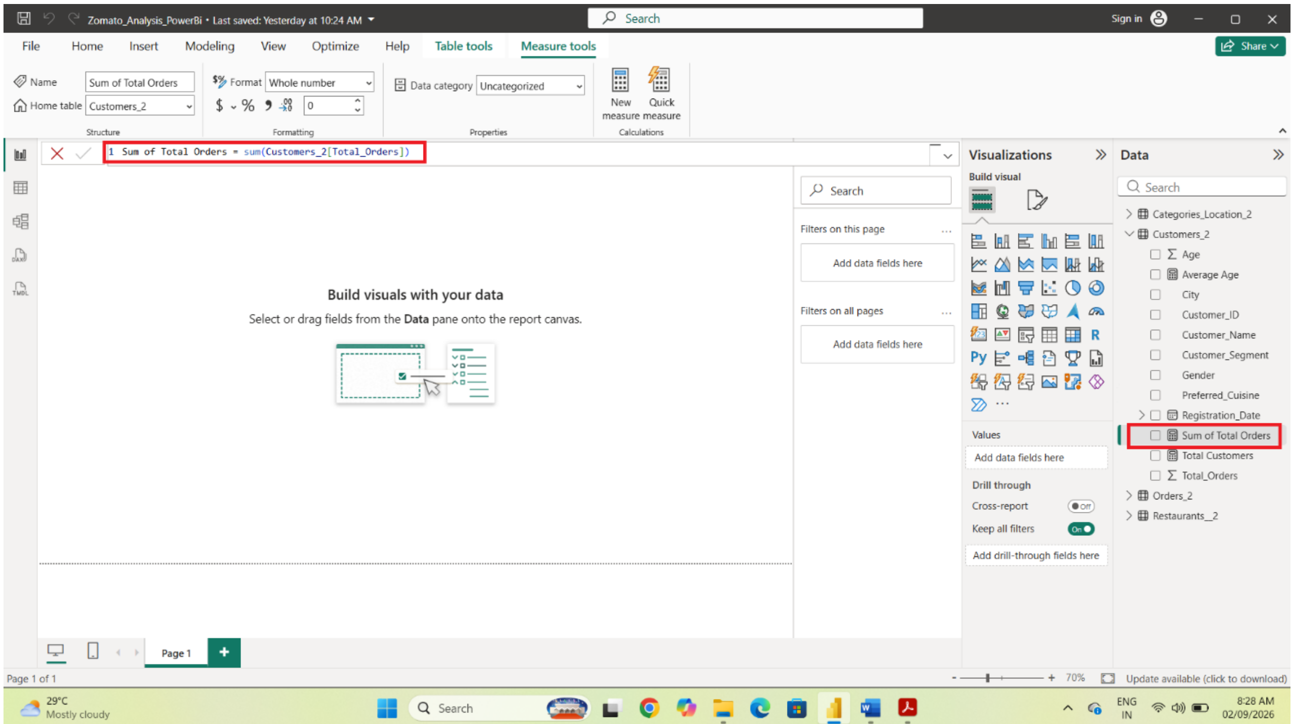
Create a distinct-count measure for customers.



Create the Customer Total Orders measure

Steps: Select the Customers table → Modeling → New measure. Create a measure that sums the Total_Orders field to calculate the total number of customer orders recorded in the customer table.

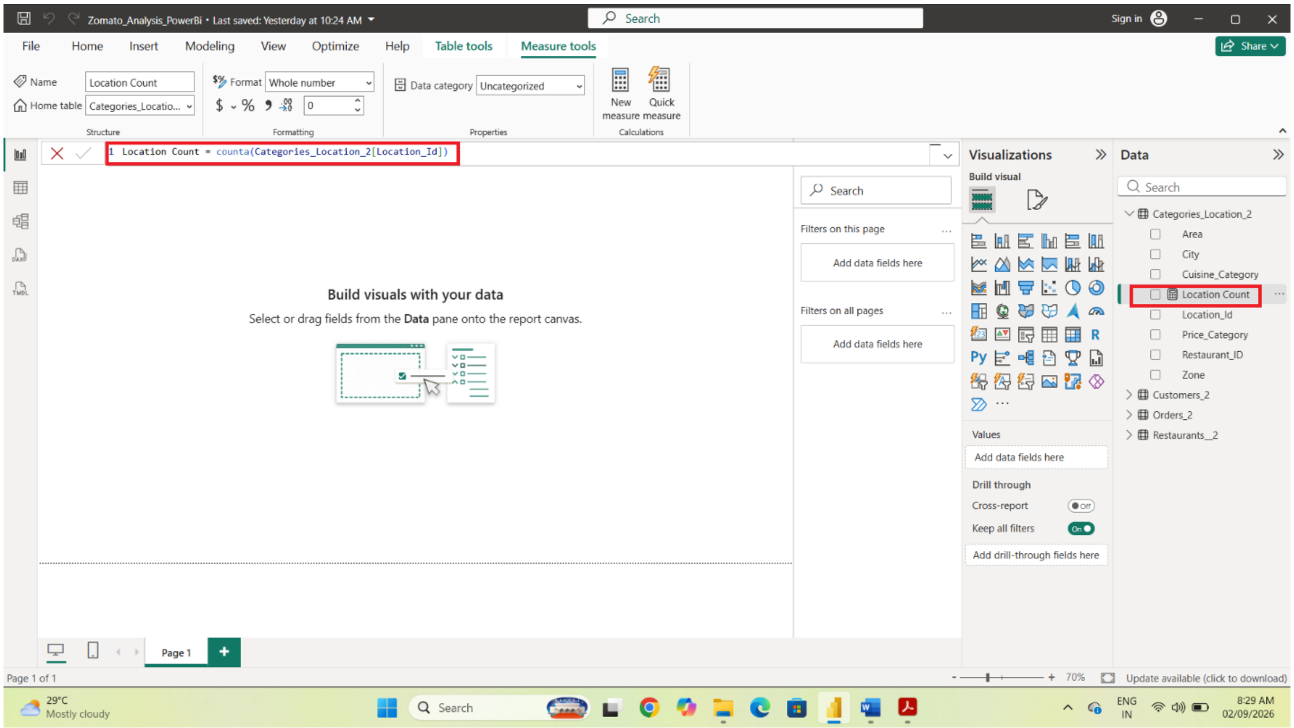
Create a measure by summing the customer Total_Orders field.



Create the Location Count measure

Steps: Select the Categories_Locations table → Modeling → New measure. Create Location Count using the appropriate location field and verify that the measure appears under the table.

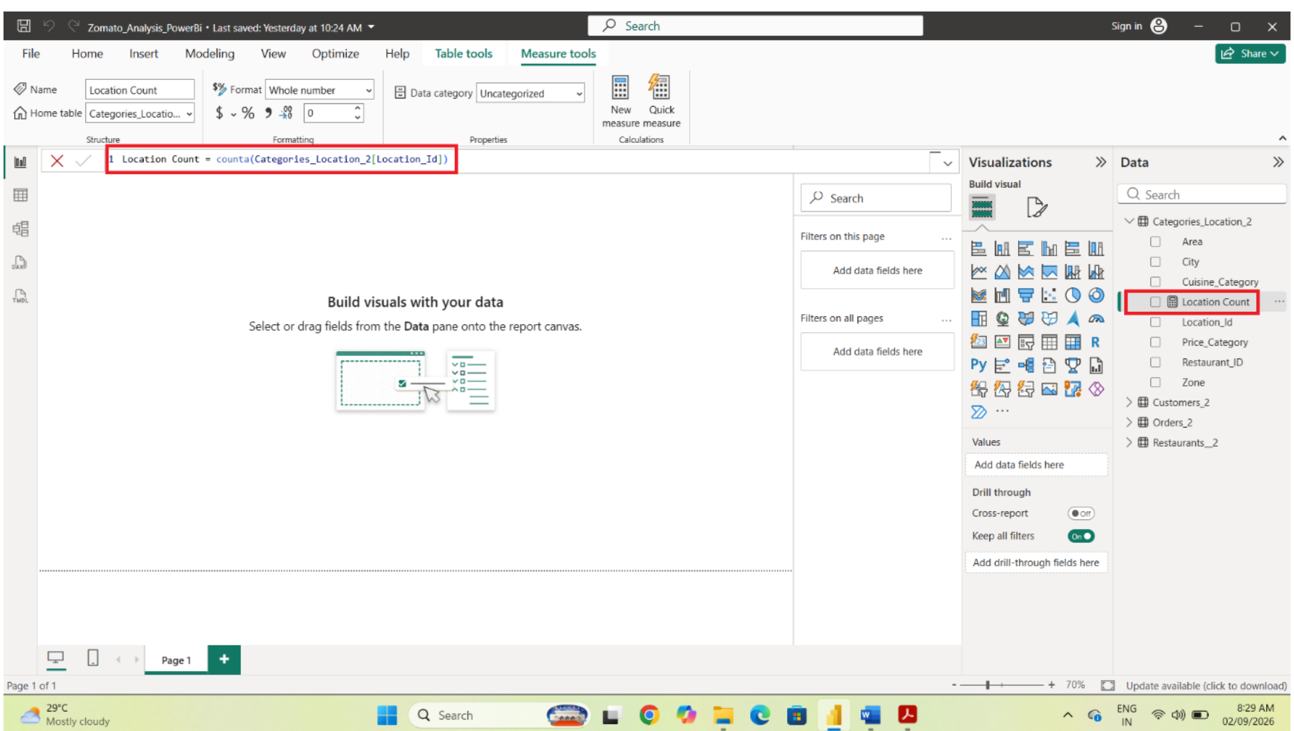
Create and verify the Location Count measure.



Verify the Location Count measure

Steps: Select the Location Count measure in the Data pane and review its formula and formatting in the Measure tools area. Confirm that the measure is ready to be used in KPI cards or location analysis.

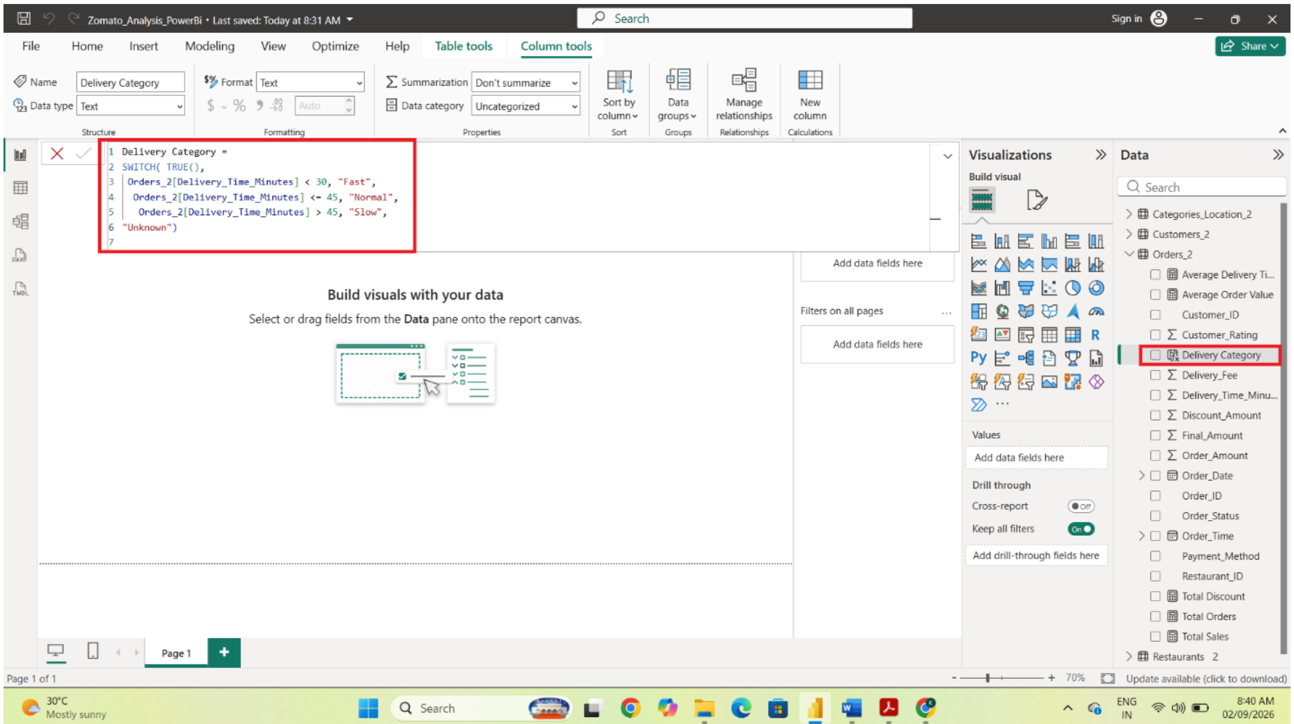
Verify the Location Count measure before using it in the dashboard.



Create the Delivery Category calculated column

Steps: Select the Orders table and choose Modeling → New column. Create Delivery Category using delivery time: Fast for less than 30 minutes, Normal for 30–45 minutes, and Slow for more than 45 minutes. Review the calculated values in the Data pane.

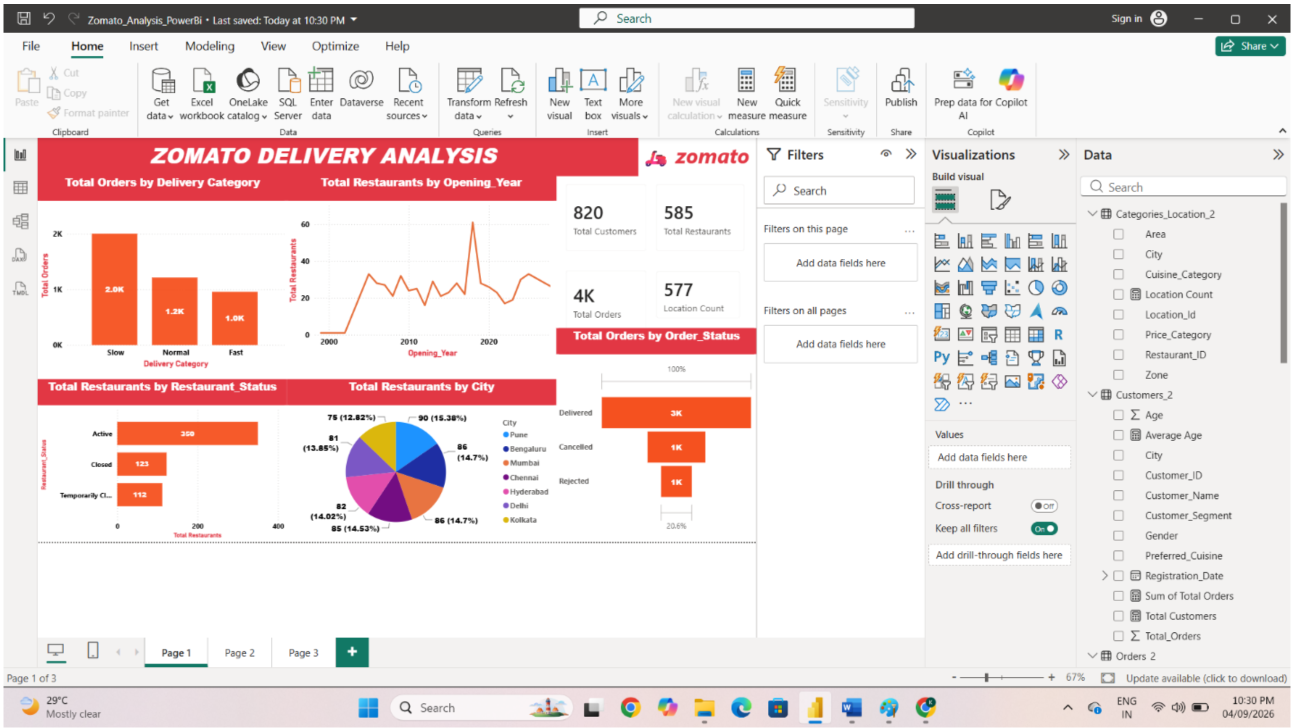
Create the Delivery Category calculated column from delivery time.



Build Dashboard Page 1

Steps: Create the first report page and add KPI cards, bar charts, a line chart, a pie chart, and a funnel chart. Use the visuals to show delivery category, restaurant opening year, restaurant status, restaurants by city, and order status.

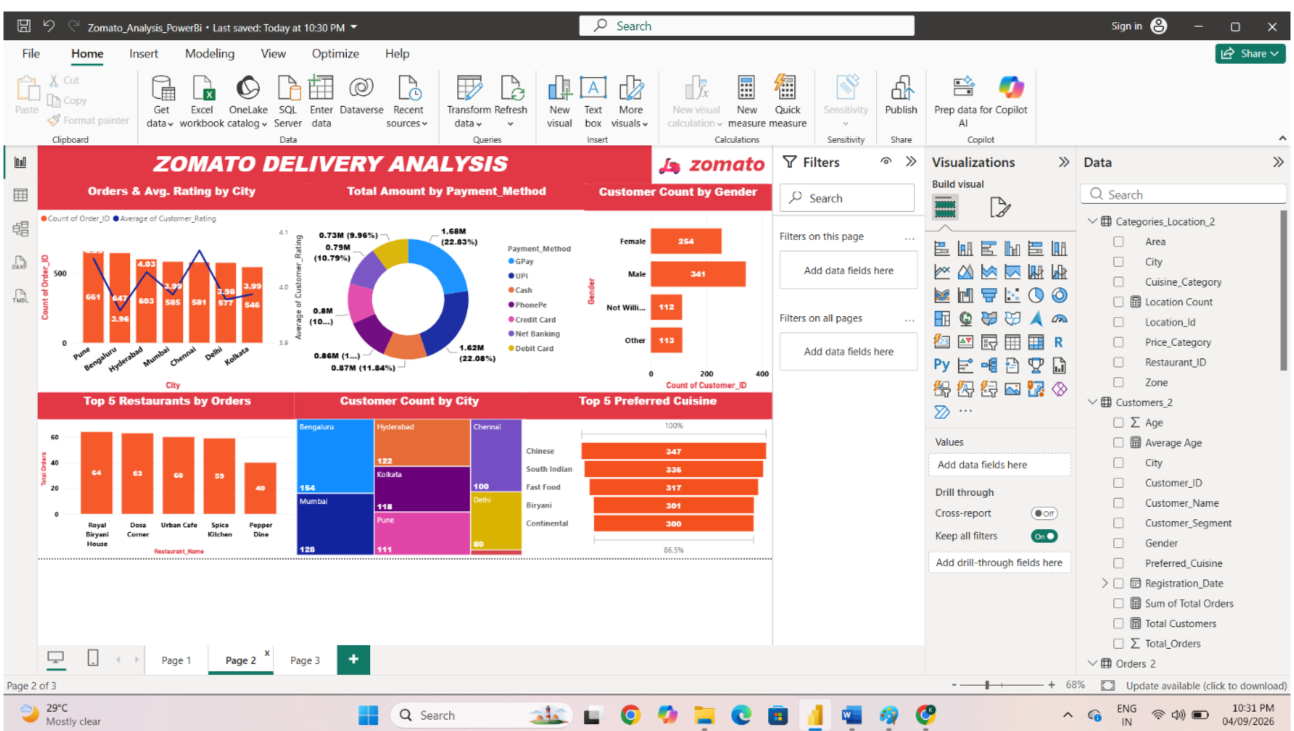
Build the first dashboard page with the overview visuals.



Build Dashboard Page 2

Steps: Create the second report page. Add a combo chart for orders and average rating by city, a donut chart for payment method, a bar chart for customer gender, a Top 5 restaurant chart, a treemap for customer count by city, and a Top 5 cuisine chart.

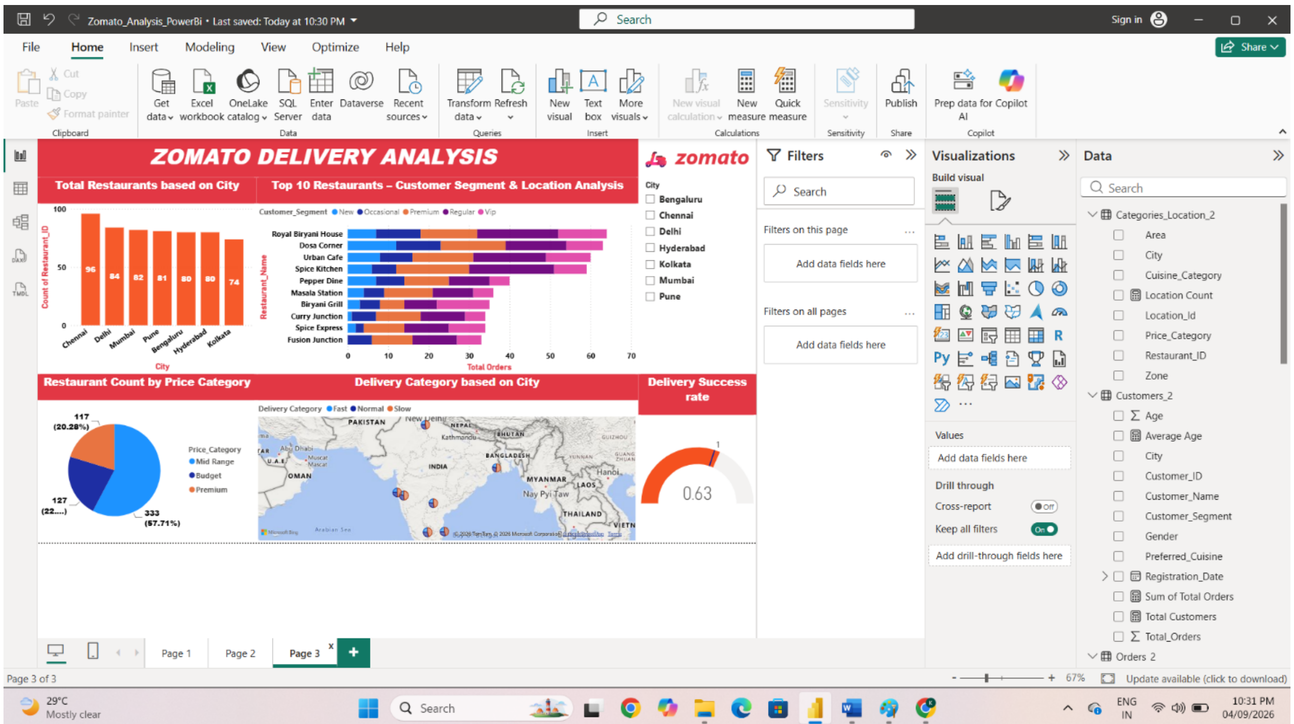
Build the second dashboard page for customer, payment, and restaurant analysis.



Build Dashboard Page 3

Steps: Create the third report page. Add city-wise restaurant analysis, the Top 10 Restaurants – Customer Segment & Location Analysis, price-category analysis, delivery category by city, and the Delivery Success Rate gauge. Add the City slicer for interactive filtering.

Build the third dashboard page for detailed restaurant and delivery analysis.



The completed Power BI report contains three dashboard pages covering delivery performance, restaurant performance, customer behavior, payment preferences, cuisine preferences, location analysis, and delivery success. The dashboards use interactive filtering and consistent visual formatting.